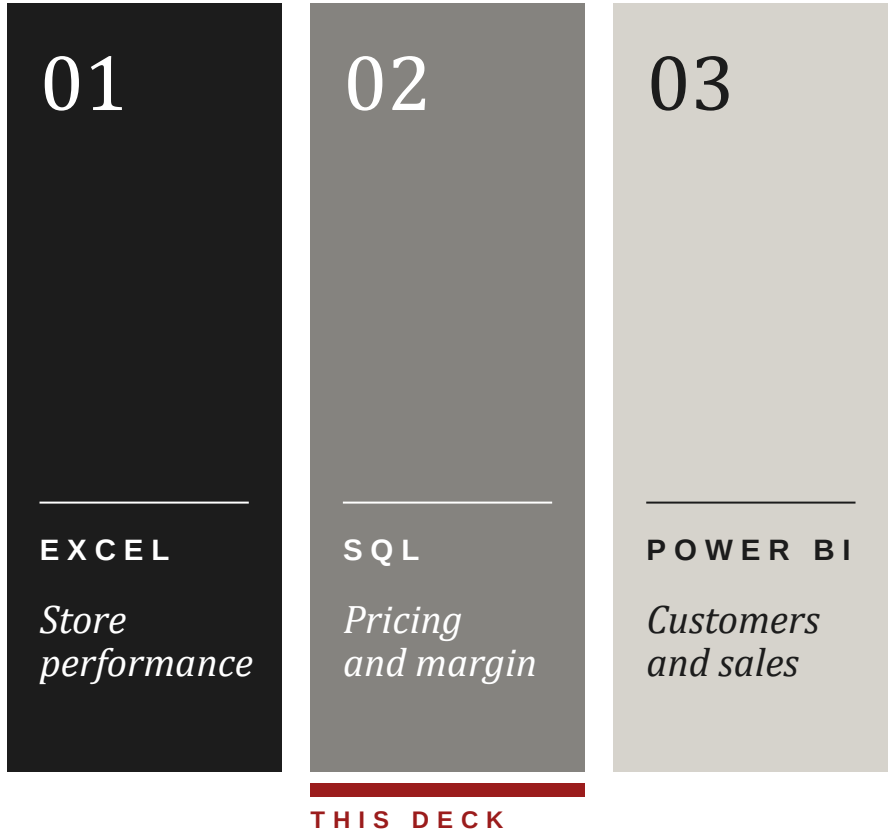
Pricing, Promotions and Margin

Problem 2: what every markdown really earns

ANMOL AGARWAL / FOR THE HEAD OF MERCHANDISING



35 STORES / 7 COUNTRIES

3.3M TRANSACTION LINES

3 PROBLEMS / ONE STORY

SIMULATED RETAIL DATASET FOR TEACHING | ALL MONEY IN USD

02

SQL / HEAD OF MERCHANDISING

How might we help Merchandising see, for every sale phase, whether it earned its margin, so that depth, timing and categories are chosen on evidence?

SITUATION

Zara sells at full price most of the year and marks down only in its Winter and Summer Sales. In 2024 the US and UK also ran Black Friday and started the Winter Sale six days early.

COMPLICATION

19.6% of sale lines are marked down but bring only 13.4% of net sales. Gross margin falls from 60.6% at full price to -40.8% at 70% off.

QUESTION

Which markdowns earn their margin, and which only lower the price of sales that would have happened anyway?

PROBLEM STATEMENT

The Head of Merchandising is facing markdowns of unknown value while trying to protect gross margin, which results in discounts that may only lower the price of sales that would have happened anyway.

How this deck follows the brief

MAP OF THE DECK

Brief part	Section (same number as in the SQL file)	What it answers	Slides
Part A	Business framing	Issue tree, causal loop, outcome map, BRD, hypotheses	2 (full note in Word)
Part B	Sections 0-1: setup	Load check, exchange rates, English country and city names	In the SQL file
Part B	Section 2: Data checks (cleaning)	Can we trust the numbers?	5-6
Part B	Section 3: Sales and margin	KPIs by year, category margin, markdown depth, what ends up in the sale	7-11
Part B	Section 4: Promotions	Sale phases, Black Friday and the early sale, monthly trend	12-15
Part B	Section 5: Products	Top products, Pareto, returns	16-20
Part B	Section 6: Reusable report	Stored procedure promo_report	21-22
Part C	Recommendations	Discount policy, Black Friday and buying decisions, RICE	23-26

The answer up front: markdowns beyond 50% destroy value

SUMMARY

-40.8%

gross margin on lines sold at 70% off (60.6% at full price)

Phase 3

of every sale loses 55-76% of normal daily margin

-14.4%

US/UK full-price sales before Christmas 2024, after Black Friday and the early sale

2x

return rate of one jean style (Z1975) against all jeans

Rules used in every query

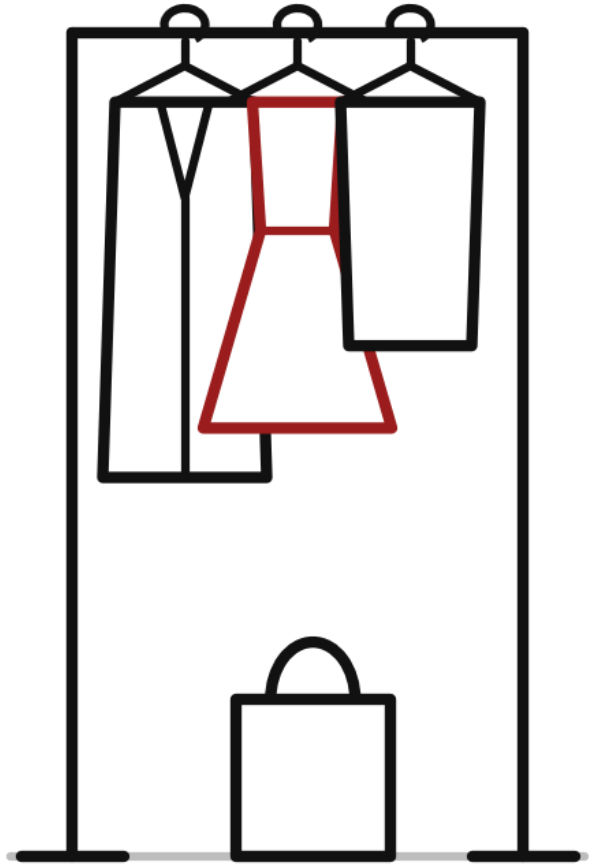
- All money in USD via exchange_rates (1 EUR = 1.08, 1 GBP = 1.27, 1 CNY = 0.14); money from Line_Total, never Invoice_Total.
- Net sales = sales + returns; returns carry a positive Quantity but a negative Line_Total, so returned units count as negative.
- Gross margin = net sales - production cost x net units; cost in EUR x 1.08, and 25 costs about 100x too high are divided by 100.

PART B | ANALYSIS IN SQL

Section 2: Data checks (cleaning)

Questions answered in this section

- Which invoices have an Invoice Total that does not match their lines?
- Which lines have no size or colour?
- Which products have a production cost that cannot be right?
- Which sales were served by an employee from another store?
- Which products were never sold - and what do we do about each issue?



Trust the numbers: six data issues found and handled

QUERIES 2.1-2.6 | Brief: Trust the numbers - wrong invoice totals, missing size/colour, wrong costs, other-store staff, never-sold products

Check	Rows affected	What we did
Invoice_Total differs from the sum of its lines	4,200 invoices	All money from Line_Total
Lines with no size	13,363	Kept; excluded only from size analysis
Lines with no colour	6,681	Kept; colour from products
Production cost about 100x too high	25 products	Cost divided by 100
Sale served by staff from another store	1,452 lines	Kept; no effect on margin
Products never sold	36	Left out of Pareto counts

TAKEAWAY

The data is usable once three rules are applied: money from Line_Total, costs above 200 EUR divided by 100, returns as negative units.

Without the cost fix, 25 products (worst: shoes at 4,469 EUR) would make margin look far lower.

MYSQL OUTPUT | result grid from MySQL Workbench

Issue_ID	Table_Name	Column_Name	Issue	Rows_Affected	Fix_Applied
1	transactions	Invoice_Total	Does not equal sum of Line_Totals; mostly not r...	4200	Not corrected. All money from Line_Total; invoic...
2	transactions	Size	Size missing, spread across all stores	13363	Lines kept; size labelled Unknown; excluded onl...
3	transactions	Color	Colour missing	6681	Colour taken from products.Color via Product_ID
4	products	Production_Cost	Cost about 100x too high (decimal-point error) i...	25	Corrected in v_products_clean: cost > 10x sub-...
5	transactions	Employee_ID	Sale served by an employee from another store	1452	Kept; no effect on sales or margin; handle in em...
6	products	Product_ID	Products never sold	36	Kept in products; excluded from products-sold a...
NULL	NULL	NULL	NULL	NULL	NULL

ZARA

PART B | ANALYSIS IN SQL

Section 3: Sales and margin

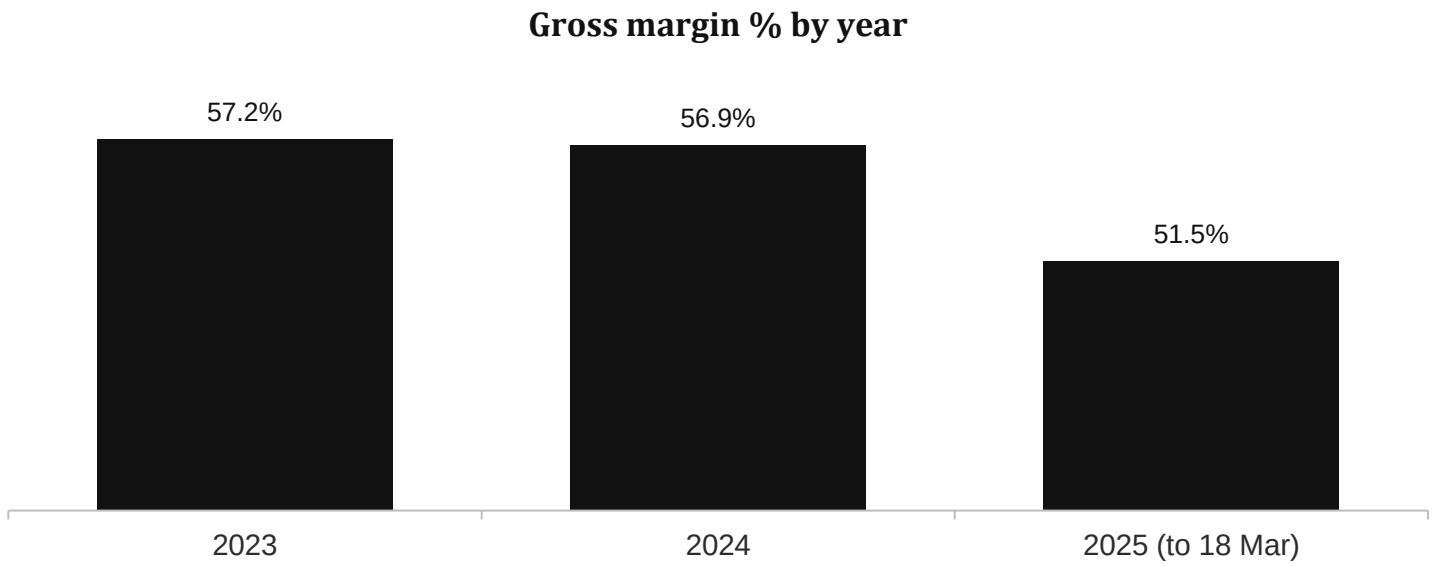


Questions answered in this section

- KPIs: gross sales, returns, net sales, gross margin and margin % by year
- Category margin: which sub-categories earn least, and why?
- Markdown depth: at which discount does Zara start losing money?
- What ends up in the sale: which collections are overbought or never discounted?

Net sales grew 1.3% in 2024, but margin % slipped

QUERY 3.1 | Brief: By year - gross sales, returns, net sales, gross margin and gross margin %



TAKEAWAY

Net sales $57.5M in 2023 and $58.2M in 2024 (+1.3%); margin % fell 0.3 points.

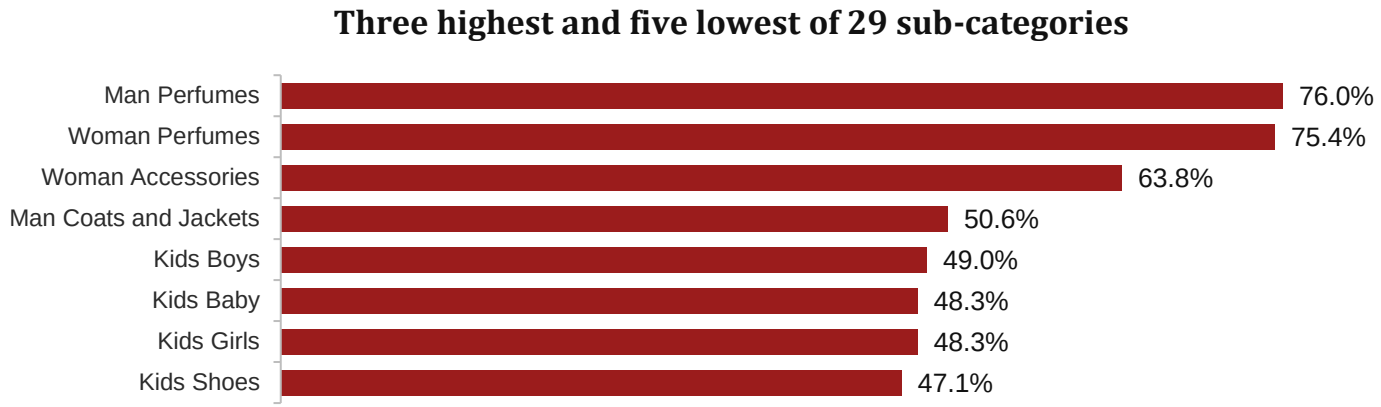
Early 2025 runs at 51.5% because January to March is Winter Sale season, so it is compared only with the same dates in 2024.

MYSQL OUTPUT | result grid from MySQL Workbench

yr	gross_sales_usd	returns_usd	net_sales_usd	gross_margin_usd	gross_margin_pct
2023	64158707	-6672380	57486327	32859083	57.16
2024	65567112	-7345895	58221218	33118728	56.88
2025	12313936	-1443775	10870161	5601100	51.53

Kids and men's coats earn least; perfumes earn most

QUERY 3.2 | Brief: Gross margin % by category and sub-category - which earn least, and why?



MYSQL OUTPUT | result grid from MySQL Workbench

Category	Sub_Category	net_sales_usd	gross_margin_usd	gross_margin_pct
Kids	Kids Shoes	1596707	752793	47.15
Kids	Girls	2728351	1317107	48.27
Kids	Baby	1490928	720729	48.34
Kids	Boys	2422836	1188229	49.04
Man	Coats and Jackets	10977590	5555140	50.60
Man	Shoes	3994025	2052239	51.38
Woman	Shoes	6727813	3484553	51.79
Woman	Coats and Trench Coats	9307904	4964446	53.34
Man	Knitwear and Sweatshirts	4149031	2243088	54.06

TAKEAWAY

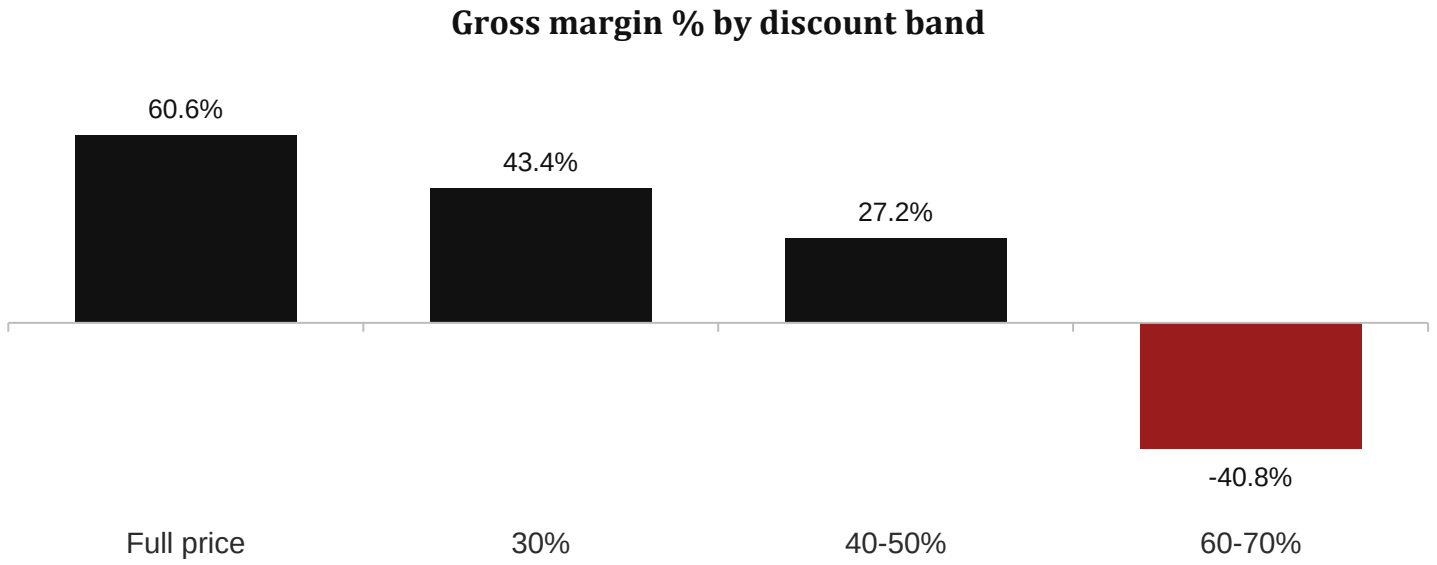
By category: Woman 57.8%, Man 56.0%, Kids 48.3%.

Kids earns least: low prices against cost and heavy discounting. Men's coats and jackets lose margin because 29% of their units sell on markdown.

The result grid lists the lowest-margin sub-categories first; perfumes (75-76%) sit at the bottom of the full list.

Margin turns negative at 70% off

QUERY 3.3 | Brief: Markdown depth bands - at which depth does Zara start losing money?



TAKEAWAY

Every step of depth costs margin: 60.6% -> 43.4% -> 27.2%.

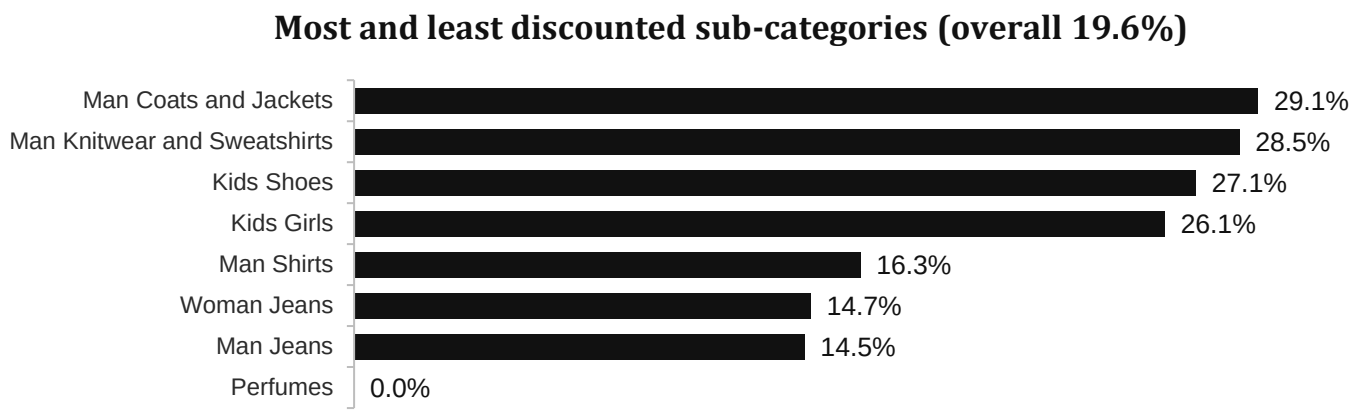
At 70% off the price is below the cost to make (-40.8%). These lines are only 1.4% of sale lines, but each one loses money.

MYSQL OUTPUT | result grid from MySQL Workbench

markdown_band	sale_lines	pct_of_sale_lines	net_sales_usd	gross_margin_pct
1 Full price	2445540	80.38	109577896	60.55
2 30%	181154	5.95	6265635	43.44
3 40-50%	373075	12.26	10127785	27.19
4 60-70%	42793	1.41	606391	-40.79

Outerwear, knitwear and kids' wear end up in the sale most

QUERY 3.4 | Brief: What ends up in the sale - which collections are overbought or never discounted?



MYSQL OUTPUT | result grid from MySQL Workbench

Category	Sub_Category	units_sold	units_on_markdown	pct_units_on_markdown
Man	Coats and Jackets	118608	34506	29.1
Man	Knitwear and Sweatshirts	118360	33720	28.5
Kids	Kids Shoes	65659	17765	27.1
Kids	Girls	154224	40286	26.1
Man	Shoes	71218	18009	25.3
Kids	Boys	131377	32895	25.0
Woman	Knitwear	151733	38000	25.0
Kids	Baby	87517	21554	24.6
Woman	Shoes	123703	28295	22.9

TAKEAWAY

About one unit in five (650,754 of 3,316,530) sells on markdown.

Coats, knitwear and kids' wear (25-29%) are overbought; jeans (about 15%) are well managed; perfumes are never discounted.

The most discounted sub-categories are also the lowest-margin ones in query 3.2: overbuying and thin margins go together.

PART B | ANALYSIS IN SQL

Section 4: Promotions



Questions answered in this section

- Sale phases: which phases are worth their margin compared with the two weeks before?
- Black Friday and the early Winter Sale (US, UK): did they add sales or move them earlier?
- Monthly trend with LAG(): how much of each peak month came from markdown?

Only the first phase of a sale pays for its discount

QUERY 4.1 | Brief: Sale phases vs the two weeks before - which phases are worth their margin?

Margin per day vs 14 days before	Phase 1	Phase 2	Phase 3
Summer Sale 2023	+20%	-23%	-57%
Winter Sale 2023-24	-15%	-56%	-76%
Summer Sale 2024	+19%	-23%	-56%
Winter Sale 2024-25 (US, UK)	+20%	-47%	-71%
Winter Sale 2024-25 (other 5)	-16%	-55%	-74%
Black Friday 2024 (US, UK)	+129%	-	-

TAKEAWAY

Phase 1 of the Summer Sales and Black Friday earn more margin per day than normal trading.

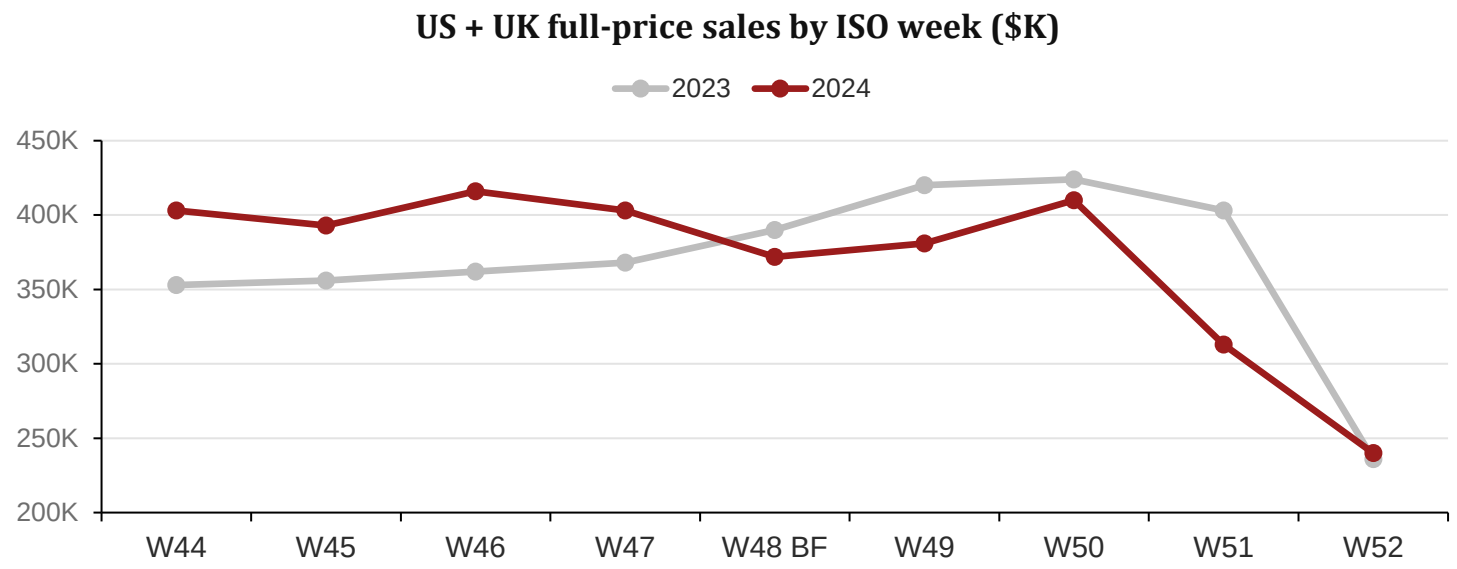
Phases 2 and 3 never pay for their discount: phase 3 loses 55-76% of normal daily margin.

MYSQL OUTPUT | result grid from MySQL Workbench

season	phase_no	countries	units	net_sales_usd	gm_pct_before	gm_pct_phase	units_per_day_change_pct	sales_per_day_change_pct	margin_per_day
Summer Sale 2023	1	All countries	78846	2622372	60.8	50.1	82	46	20
Summer Sale 2023	2	All countries	61636	1864708	60.8	45.1	42	4	-23
Summer Sale 2023	3	All countries	45457	1260914	60.8	37.1	5	-30	-57
Winter Sale 2023-24	1	All countries	81839	3101058	59.5	46.4	45	9	-15
Winter Sale 2023-24	2	All countries	54278	1835141	59.5	41.0	-4	-36	-56
Winter Sale 2023-24	3	All countries	44516	1292756	59.5	32.1	-21	-55	-76
Summer Sale 2024	1	All countries	80246	2582624	60.6	50.1	79	44	19
Summer Sale 2024	2	All countries	63247	1850057	60.6	45.4	41	3	-23

Black Friday pulled US/UK full-price sales forward

QUERY 4.2 | Brief: Black Friday and the early sale - did they add sales or move them earlier?



TAKEAWAY

Before Black Friday US/UK full-price sales ran 9-15% above 2023; after it they fell below 2023 (weeks 49-51).

In 4-24 Dec, US/UK full price fell 14.4% while the other countries grew 2.8%: sales moved earlier at a lower price.

QUERY OUTPUT | calculated from the dataset with the same logic as the query

region	full_price_2023	full_price_2024	full_price_change_pct	markdown_2023	markdown_2024	total_change_pct	gm_pct_2023	gm_pct_2024
US + UK	1,247,641	1,068,097	-14.4	0	273,727	7.5	60.9	57.1
Other 5 countries	3,201,309	3,289,992	2.8	0	0	2.8	59.8	59.9

December peaks; January and July run on markdown

QUERY 4.3 | Brief: Monthly trend with LAG() - how much of each peak came from markdown?



TAKEAWAY

December is the peak, and in 2024 more of it (22% vs 15%) came from markdown.

January and July are mostly markdown and run at 40-47% margin instead of about 60%. Nov 2024 was +12.8% year on year (Black Friday).

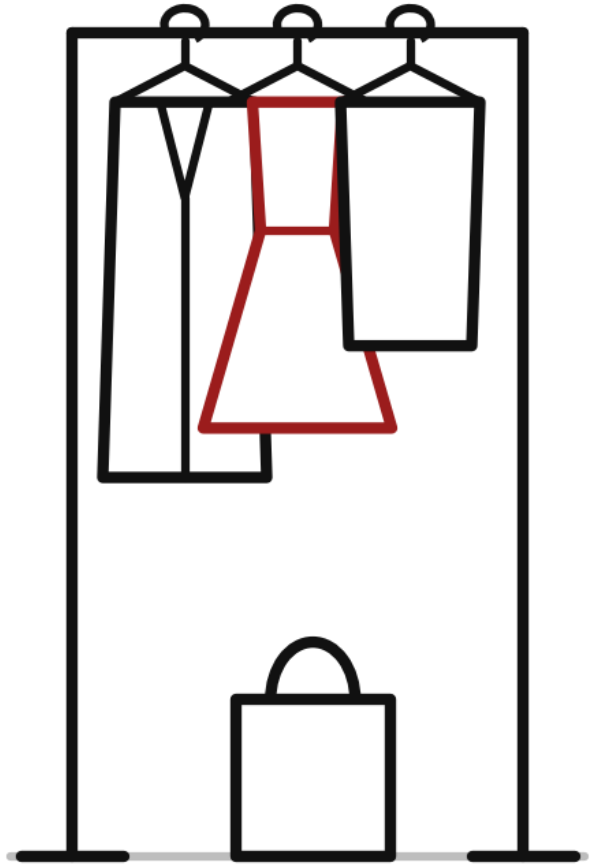
QUERY OUTPUT | calculated from the dataset with the same logic as the query

month	net_sales_usd	gm_pct	markdown_share_pct	mom_pct	yoy_pct
2023-12	7,177,744	57.1	15.1	32.6	NULL
2024-01	4,154,297	41.5	58.9	-42.1	-16.0
2024-07	4,345,067	47.0	52.8	2.4	-8.7
2024-11	6,108,561	59.3	3.7	0.8	12.8
2024-12	7,394,841	55.9	22.4	21.1	3.0
2025-03 (partial)	2,903,256	60.2	0.0	NULL	NULL

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PART B | ANALYSIS IN SQL

Section 5: Products



Questions answered in this section

- Top 10 products by net sales in each category (RANK)
- Pareto: what share of products makes up 80% of net sales?
- Returns: return rate by sub-category, size, country and price type, and which products are outliers

A few hero coats and knits lead each category

QUERY 5.1 | Brief: Top 10 products by net sales in each category (RANK)

Category	No. 1 product	No. 1 sales	No. 10 sales
Woman	1817 ZW Collection Puffer Coat (white)	1.72M	0.41M
Man	803 Basic Cardigan (red)	1.17M	0.43M
Kids	2088 Knit Skirt (camel)	0.35M	0.11M

- Woman no. 2: 4966 Oversize Blazer (1.38M).
- Man no. 2: 3445 Technical Overcoat (0.99M).
- Kids no. 2: 3621 Velcro Trainers (0.22M).

TAKEAWAY

Woman depends most on a few hero products: its no. 1 sells four times its no. 10.

The leaders are coats and knitwear, the same collections that end up overbought (query 3.4).

MYSQL OUTPUT | result grid from MySQL Workbench

Category	rank	Product_ID	Description_EN	Color	Sub_Category	net_sales_usd	units
Kids	1	2088	KNIT SKIRT	CAMEL	Girls	350835	28299
Kids	2	3621	VELCRO TRAINERS	NAVY BLUE	Kids Shoes	222896	6331
Kids	3	722	CARGO SHORTS	GREEN	Boys	198845	16290
Kids	4	5072	COTTON ROMPER	LIGHT BLUE	Baby	173672	13719
Kids	5	3963	CHUNKY TRAINERS	CAMEL	Kids Shoes	165921	7281

A quarter of the range makes 80% of net sales

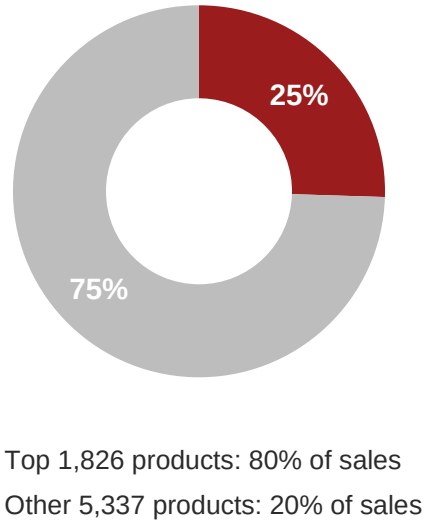
QUERY 5.2 | Brief: Pareto - what share of products makes up 80% of net sales?

25.5%

of sold products make 80% of net sales

1,826

of 7,163 products sold (7,199 in the catalogue)



TAKEAWAY

Close to the classic 80/20 pattern.

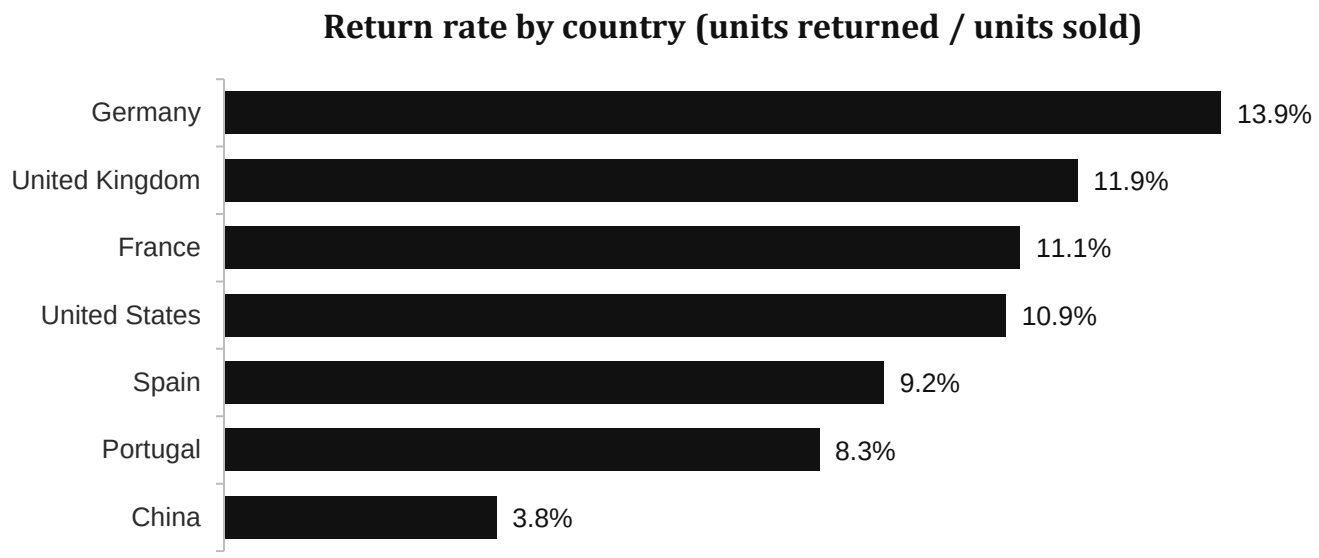
Buying depth should follow the top quarter of styles; the long tail is where markdown risk sits.

MYSQL OUTPUT | result grid from MySQL Workbench

products_making_80pct	products_sold	pct_of_sold_products	products_in_catalogue	pct_of_catalogue
1826	7163	25.5	7199	25.4

Germany returns four times as much as China

QUERY 5.3 | Brief: Return rate by sub-category, size, country and full price vs markdown



TAKEAWAY

Fitted clothing (dresses, jeans) comes back most, and size 36 is returned most (21.4%).

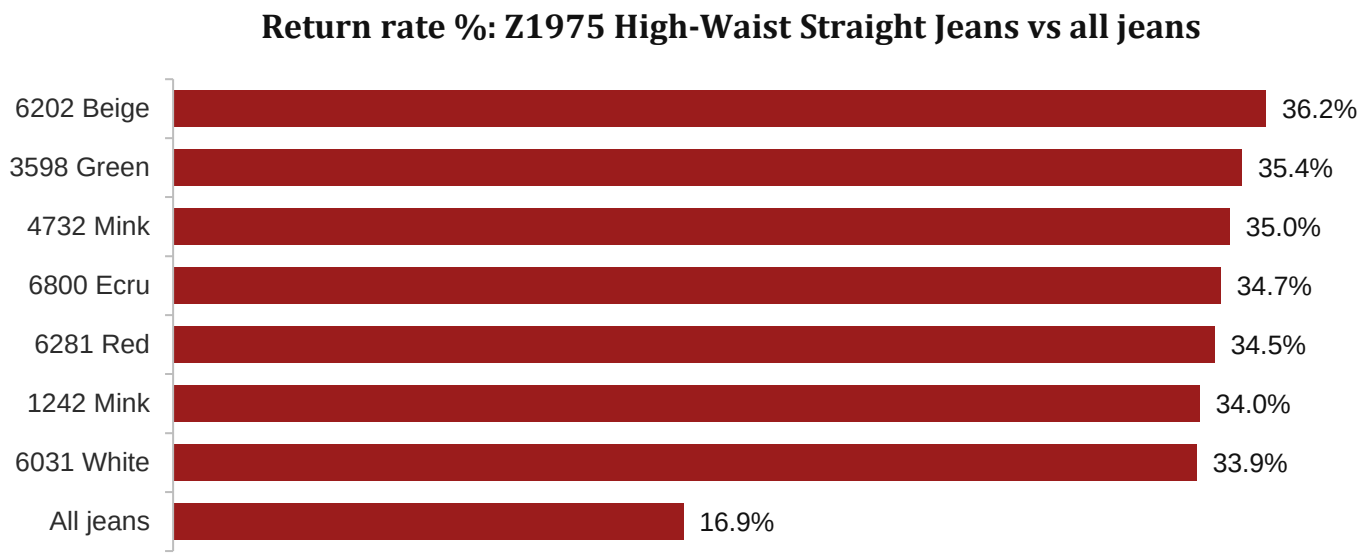
Markdown items are returned less than full-price ones (6.6% vs 10.6%), so returns are a fit problem, not a sale problem.

QUERY OUTPUT | calculated from the dataset with the same logic as the query

breakdown	value	units_sold	units_returned	return_rate_pct
sub-category	Dresses	213,335	37,123	17.4
sub-category	Jeans	303,727	51,369	16.9
sub-category	Perfumes	112,252	1,126	1.0
size	36	63,188	13,524	21.4
country	Germany / China	439,026 / 459,039	61,038 / 17,454	13.9 / 3.8
price type	full price / markdown	2,665,776 / 650,754	282,140 / 42,745	10.6 / 6.6

One jean style is returned twice as often as all jeans

QUERY 5.3b | Brief: Products whose return rate is far above their sub-category - what is wrong?



TAKEAWAY

All 7 outliers are the same cut in 7 colours, so the problem is the fit or size chart of the Z1975 pattern, not a colour or batch.

Bringing it to the jeans average would avoid about 1,600 returned units.

MYSQL OUTPUT | result grid from MySQL Workbench

Product_ID	Description_EN	Color	Sub_Category	units_sold	return_rate_pct	sub_category_rate_pct	times_sub_category
6202	Z1975 HIGH-WAIST STRAIGHT JEANS 8	BEIGE	Jeans	1340	36.2	16.9	2.1
3598	Z1975 HIGH-WAIST STRAIGHT JEANS 1	GREEN	Jeans	1249	35.4	16.9	2.1
4732	Z1975 HIGH-WAIST STRAIGHT JEANS 3	MINK	Jeans	1273	35.0	16.9	2.1
6800	Z1975 HIGH-WAIST STRAIGHT JEANS 4	ECRU	Jeans	1256	34.7	16.9	2.1
6281	Z1975 HIGH-WAIST STRAIGHT JEANS 7	RED	Jeans	1421	34.5	16.9	2.0

PART B | ANALYSIS IN SQL

Section 6: Reusable report



Questions answered in this section

- Stored procedure promo_report(start, end, country): sales, units and margin against the period before

promo_report: any market can test its own promotion

QUERY 6 | Brief: Stored procedure promo_report(start, end, country) vs the period before

```
CALL promo_report('2024-11-29', '2024-12-02', 'United States');
```

+310% net sales vs the 4 days before

+421% units (9,367 vs 1,799)

+242% margin (201K vs 59K)

TAKEAWAY

Black Friday quadrupled US sales in its four days, at a 10-point lower margin (51.4% vs 61.6%).

Inputs: start date, end date, country ('All' for every country). Merchandising can now judge any promotion without a new analysis.

MYSQL OUTPUT | result grid from MySQL Workbench

period_start	period_end	prev_start	prev_end	net_sales_usd	prev_net_sales_usd	sales_change_pct	units	prev_units	units_change_pct	margin_usd	prev_margin_usd	margin_chan
2024-11-29	2024-12-02	2024-11-25	2024-11-28	391207	95430	309.9	9367	1799	420.7	201264	58803	242.3

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PART C

Recommendations



Questions answered in this section

- Discount policy by category: maximum depth, how often, which categories rarely go on sale
- Keep Black Friday and the early Winter Sale? How to change buying for overbought collections, and the margin each change protects
- Priority of every recommendation with a RICE score

Cap markdown at 50%, and at 40% for Kids

DISCOUNT POLICY BY CATEGORY

Category	Margin % at full / 30 / 40 / 50 / 70% off	Maximum depth	How often	Rarely or never discount
Woman	61.2 / 43.5 / 34.1 / 20.8 / -	50%	Winter and Summer Sales, phases 1-2	Perfumes (never), jeans, blazers
Man	60.5 / 43.5 / 32.7 / 19.0 / -37.0	50% (stop the 70% phase)	Winter and Summer Sales, phases 1-2	Perfumes (never), jeans, shirts
Kids	55.2 / 36.5 / 25.4 / 10.7 / -48.9	40%	Same; 50% only for end-of-season clearance	-

≈ $220K

margin protected a year by ending the 70% phase and capping Kids at 40% (if the same units sell)

TAKEAWAY

The 70% lines lost $247K of margin; sold at 50% they would have earned $157K.

Kids at 40% instead of 50% lifts margin from $43K to $124K.

Keep Black Friday, end the early Winter Sale, buy less

BLACK FRIDAY AND OVERBOUGHT COLLECTIONS

1

Keep Black Friday, but only on overbought lines

It nearly doubled US/UK margin in its four days (+$164K vs 2023), but full-price sales in the next two weeks fell 6.7% while other countries grew 4.3%.

2

Move the US/UK Winter Sale back to 26 Dec

Starting on 20 Dec added sales at a 12-point lower margin (48.9% vs 60.9%) and did not lift Nov-Jan margin above US/UK's own 11% growth.

3

Buy about 10% less of four collections

Coats and jackets, knitwear, kids shoes and girls' wear gave away $3.58M of discount. Cutting their markdown share to the 19.6% average saves about $0.5M a year.

Priority: cap depth first, then fix the calendar, then buying

RICE PRIORITISATION

Recommendation	Reach / year	Impact	Confidence	Effort (weeks)	RICE	Priority
Cap depth: no 70% phase; Kids max 40%	36,337 lines	2	60%	2	21,800	1
Black Friday on overbought lines only; Winter Sale back to 26 Dec	30,632 lines	1	50%	1	15,300	2
Buy less of the four overbought collections	57,182 units	2	50%	8	7,100	3

Score = Reach x Impact x Confidence / Effort. Impact: 3 = massive, 2 = high, 1 = medium.

Confidence is below 100% because the data has no stock levels and only one Black Friday, and a buying change takes a full season to show.